

Theories of Computation: Summative Assignment 1

To be handed in on Canvas before **Tuesday 7th March, 12pm**

Exercise 1 Let $\Sigma = \{a, b\}$ be the alphabet for this question. Let E be the regular expression aa^*bb^* .

1. Are the following words recognized by E ? [3 marks]

- (a) ε
- (b) aaa
- (c) $abbb$

2. Give a DFA with at most five states that accepts the language described by E . [3 marks]

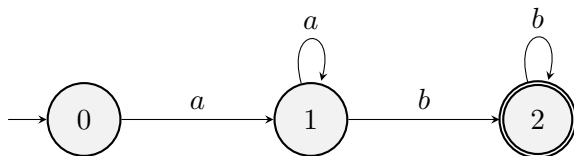
NB. It can be a partial DFA.

3. Give a context-free grammar that generates the same language. [3 marks]

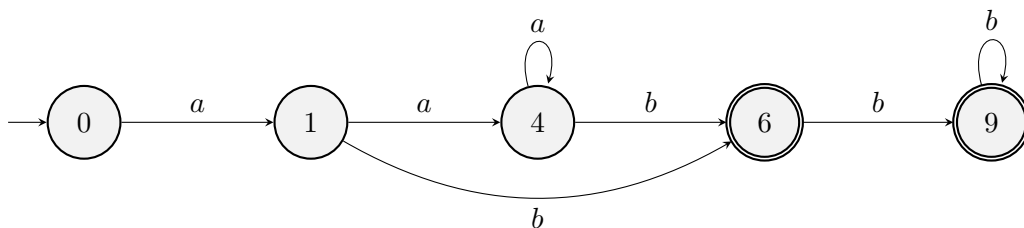
Solution 1 1. The language described by this regular expression is $L = \{a^n b^m \mid n, m \geq 1\}$

- (a) ε is not recognized.
- (b) aaa is not recognized.
- (c) $abbb$ is recognized.

2. The minimal solution is:



If one follows the procedure given in the handout to go from regular expressions to DFA, one gets:



3. Since the language is regular, we can directly obtain from the DFA the following grammar that represents it.

$$\begin{aligned} \Rightarrow S_0 &::= aS_1 \\ S_1 &::= aS_1 \mid bS_2 \\ S_2 &::= bS_2 \mid \varepsilon \end{aligned}$$

Exercise 2 Let $\Sigma = \{a, b\}$ be the alphabet for this question. Consider the following context-free grammar G :

$$\begin{aligned} \Rightarrow S &::= aSa \mid B \\ B &::= BB \mid b \end{aligned}$$

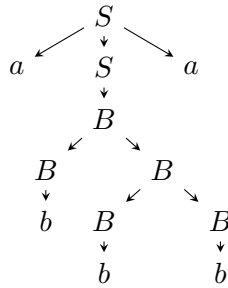
1. Give a derivation tree for the word $abbba$ and the corresponding leftmost derivation. [3 marks]

2. What is the language $L(G)$ generated by this grammar? [3 marks]

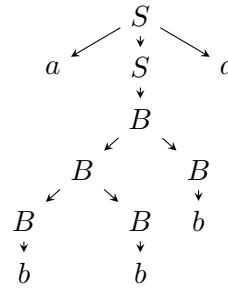
3. Is $L(G)$ regular? Justify your answer.

[3 marks]

Solution 2 1. There are two possible answers to this question:



$$\begin{aligned} S &\Rightarrow aSa \Rightarrow aBa \Rightarrow aBBa \Rightarrow \textcolor{red}{abBa} \\ &\Rightarrow abBBa \Rightarrow abbBa \Rightarrow abba \end{aligned}$$

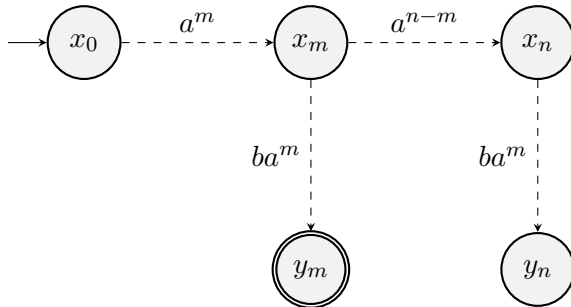


$$\begin{aligned} S &\Rightarrow aSa \Rightarrow aBa \Rightarrow aBBa \Rightarrow \textcolor{red}{aBBBa} \\ &\Rightarrow abBBa \Rightarrow abbBa \Rightarrow abbbba \end{aligned}$$

2. $L(G) = \{a^n b^m a^n \mid n \geq 0, m \geq 1\}$

3. $L(G)$ is not regular.

Suppose that we can give a deterministic finite automaton (DFA) D that recognizes L_1 . For any $n \in \mathbb{N}$, let x_n denote the state of D reached after reading a^n . A given state x_n accepts the word ba^n , but not the word ba^m for any m such that $1 \leq m < n$, while ba^m would be accepted by the corresponding state x_m .



Hence any x_n is inequivalent to all x_m for $m < n$. This means that the DFA D would need to consist of infinitely many different states, which contradicts its assumed finiteness.

Exercise 3 There are uncountably many colours in the ideal visible light spectrum. A computer science student had the idea to use the RGB gamut as a base to design a grammar to describe any colour as a string of: a number of R symbols, followed by a number of G symbols, followed by a number of B symbols. Will they succeed in describing all the colours? Justify your answer. [2 marks]

Solution 3 This is impossible because the grammar can only generate a countable number of finite strings while the set of colours is assumed to be uncountable.